

Designing an AIGC-Supported Aesthetic Learning Community for Art learners: Toward Equitable Art Education Access

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Abstract

In China, significant disparities in art education quality persist between urban and rural regions, with this imbalance having compromised educational equity and constrained possibilities for individual diversified growth. While the internet provides opportunities for the public to browse art works, art education requires practical guidance, which demands experienced mentors to offer immediate feedback and support. This is often challenging in resource-scarce regions. To solve this problem, we have tailored an AIGC(AI-Generated Content)-based aesthetic education community platform framework for students studying art. The platform features core modules centered around "inspiration generation—creative practice—sharing and collaboration": it generates creative sketches and recommends art information through model analysis of user input; the step-by-step drawing guidance module based on CycleGAN offers sequential teaching during stages such as sketching, coloring, and detailing; it also supports work sharing and community interaction. The platform provides high-quality resources for learners by offering accessible and personalized art guidance, promoting educational equity.

CCS Concepts

• Human-centered computing → HCI design and evaluation methods.

Keywords

AI-assisted creativity; Educational resource distribution; Aesthetic education

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1 Introduction

As a crucial pathway for fostering creativity development, cultural awareness, and social identity formation among adolescents, the inequitable distribution of art education resources has become an increasingly prominent social concern [5]. Art education fundamentally relies on systematic practical training and guidance from experienced instructors [4]. In China, urban students typically enjoy access to comprehensive art curricula and professional instruction, with numerous specialized art training institutions concentrated in major metropolitan areas [11]. However, adolescents in rural and economically disadvantaged regions frequently encounter multifaceted challenges including persistent shortages of qualified teachers, insufficient teaching materials, and a pronounced lack of personalized feedback mechanisms. Even when educational resources are nominally available, students often struggle to obtain timely and effective learning support. This persistent resource disparity not only constrains the development of individual artistic potential but also exacerbates existing barriers to social mobility.

In recent years, intelligent technologies have brought new possibilities to art education [1]. For example, tools like Midjourney and Stable Diffusion can assist in visual creation and generate creative sketches, but they often focus on content generation itself, lacking guidance and personalized cultivation of user behavior during the creative process. Existing research has mostly concentrated on the digital dissemination of art resources, with limited attention paid to process-oriented educational needs. Particularly in resource-scarce regions, where access to professional art instruction is limited, students struggle not only with inspiration but also with execution, refinement, and feedback. This highlights the need for systems that support learners throughout the full cycle of artistic creation—what can be described as the "perception—creation—communication" model.

To meet this need, two pedagogical mechanisms are especially critical: multidimensional evaluation, which assesses student work across dimensions like composition, proportion, lighting, and detail; and distributed guidance, which provides stage-specific, personalized feedback as learners progress. Yet most current AIGC tools lack these capabilities and are ill-suited for educational deployment.

Our proposed system introduces an AIGC-based framework designed to integrate these two features into a full-cycle learning process. By drawing from educational design theory and human-computer interaction research, we aim to investigate how AI can serve not just as a content generator, but as an adaptive co-instructor that supports equitable and structured art learning. This leads us to the following research questions:

- (1) How can AI be used to provide professional, multi-dimensional evaluations of students' artwork based on elements of art creation (such as composition, proportion, color usage, etc.)?
- (2) In the absence of professional teacher guidance, how can AIGC technology offer distributed guidance?
- (3) How can AIGC technology be deeply integrated into the "perception—expression—communication" learning process, providing appropriate feedback and guidance at different stages?

2 Related Works

2.1 Creation feedback and guidance supported by artificial intelligence technology

The enhancement of artistic and aesthetic abilities is increasingly recognized to follow a "perception—creation—communication" learning model [8]. Their research demonstrated that structured activities covering all three stages significantly improve students' skills. In the context of AIGC, researchers validated this model by integrating AI into each stage, supporting learners from inspiration to final presentation [10]. However, current AI tools such as Stable Diffusion mainly focus on static image generation and lack dynamic [12], interactive support during the creative process. Therefore, how to deeply embed AI into the "perception—expression—communication" learning process and provide appropriate feedback at different stages remains a critical challenge.

2.2 Exploration of intelligent aesthetic education under the perspective of educational equity

Educational resource disparities have long existed between urban and rural areas as well as across different regions, with this issue being particularly pronounced in the field of aesthetic education [9]. Existing studies have attempted to mitigate this problem through remote online classes and MOOCs (Massive Open Online Courses) [2]. However, due to a lack of interactive guidance, instructional support, and immediate feedback, the effectiveness of these approaches has been limited. In recent years, the academic community has begun to explore how AI technologies can help bridge the educational divide, such as by providing personalized learning paths for students in resource-scarce areas through intelligent tutoring systems [6]. Although some progress has been made in both technological and pedagogical aspects, there is still no systematic solution

that integrates the complete aesthetic education process of "discovery—creation—communication" into an AI-supported framework.

3 Method

To gain a clearer understanding of users' needs and behaviors in learning art, this study recruited 16 high school art students from the art class at Linbaixin High School in Shantou City, Guangdong Province. The participants ranged from beginners to those with prior painting experience, ensuring a diverse representation of learning needs across different skill levels. Before the interviews began, each participant was briefed on the purpose and intent of the study. Each interview lasted approximately 15 minutes. All interviews were audio-recorded and transcribed into text. Selected representative responses from participants are summarized below:

- (1) **Whether you have been exposed to auxiliary painting learning software before?** Most students stated that they have never used such software. A few who have used it mentioned that the main functions of the software they have used so far are to provide an art resource library and 3D modeling features to assist students in understanding. Three users reported using some creative AI drawing software online, which involves collaborative creation of aesthetically pleasing patterns through board drawing. In addition, some individuals noted that they had not seen systems before that could guide students during the drawing process; such systems are also relatively rare on the market.
- (2) **Identify the difficulties in painting?** 87.5% of the students reported feeling a lack of confidence during the painting process, identifying the initial shaping and later rubbing and modeling stages as particularly challenging. A small number of students mentioned that they sometimes struggle to identify their own issues while painting. Additionally, some students noted that when they are in a state of confusion and the teacher does not appear promptly, this situation can make them feel even more distressed.
- (3) **Whether you have a clear understanding of the picture problem?** Most students express having a fairly clear understanding of their issues and flaws in their artwork, but they do not know how to correct them. If they do not receive guidance or insights from teachers over a long period, they may easily fall into confusion, making it difficult to achieve substantial breakthroughs and progress. However, a few students still indicate a lack of self-awareness regarding their painting skills.
- (4) **What impressed me most during the painting process?** When you make mistakes in painting and need to constantly modify, the sense of achievement of painting a good picture will impress you deeply. Some students say that they do not know how to paint and feel powerless when they have no sense of direction.
- (5) **Key factors that effectively help learning to draw?** Correct guidance and teaching methods, mastery of painting skills, recognition of their own problems and correction, good learning and aesthetic accumulation.

Through interviews with 16 art students from high school in Shantou City, China, we identified significant behavioral gaps and

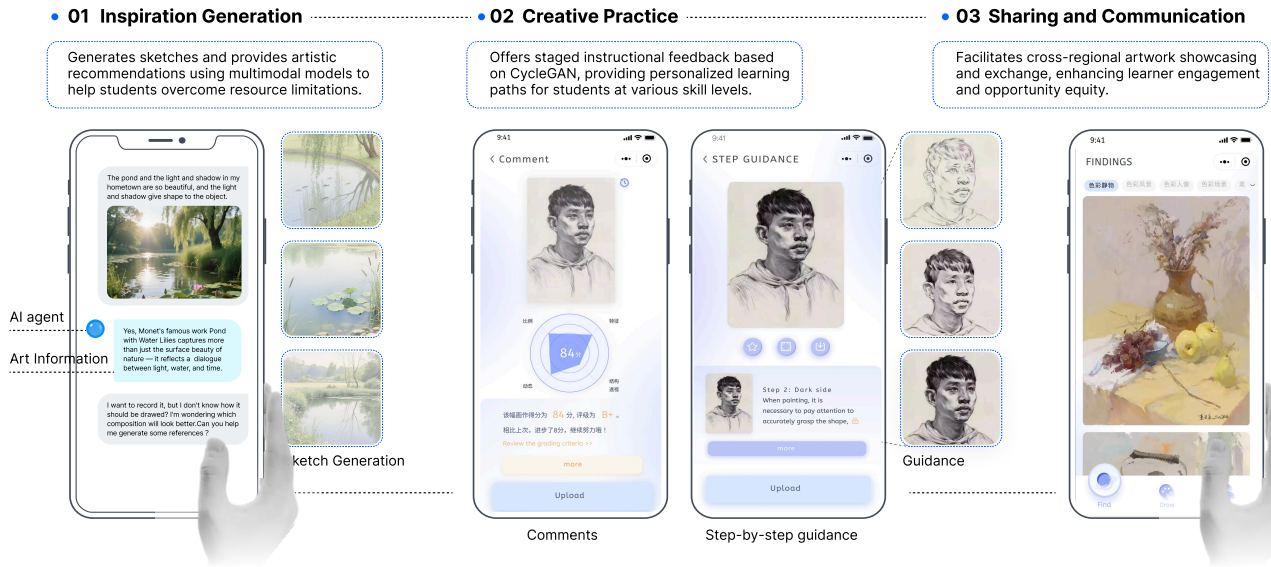


Figure 1: System function framework display.

cognitive differences in current art education, and explored the potential of AIGC-assisted tools to promote educational equity. First, students without a foundation in artistic creation, despite having a high frequency of daily practice (1-2 hours per day), often find themselves in a dilemma where they can "identify problems but cannot solve them" due to the lack of immediate feedback. This not only affects learning efficiency but also it may cause beginners to lose confidence due to long-term lack of effective guidance. Therefore, how to provide targeted guidance during stages such as inspiration generation and creative practice, offering directional references, is a key issue addressed in this paper. Additionally, how to provide students with references and point out the strengths and areas for improvement in their paintings both during and after the process, helping them enhance and progress, is another important consideration.

4 System Design

The platform is built around the complete learning process of "inspiration stimulation—creative practice—sharing and communication," integrating AIGC-assisted creation into art education scenarios. The aim is to address the disparities in art educational resources through a systematic learning process. The inspiration stimulation module combines multimodal models to generate creative sketches and artistic recommendations, helping students in resource-scarce areas break through material limitations. The step-by-step drawing guidance module uses CycleGAN for phased teaching feedback, providing personalized learning paths for students at different levels. The community interaction mechanism promotes cross-regional display and exchange of works, enhancing fairness in learning opportunities. The entire design aims to promote educational equity, exploring inclusive aesthetic education pathways empowered by AIGC.

4.1 Inspiration module

Inspiration is the first step in artistic creation and often the most challenging phase for many beginners, especially in areas with scarce educational resources where students frequently lack access to rich visual materials, examples of artistic styles, or creative guidance. To address this, our platform has designed an inspiration module based on a multimodal content understanding model. This module aims to help users break through the limitations of their creative starting point through AI-generated and recommended mechanisms. The module supports user input of textual descriptions, sketches, images, or other forms of creative cues, generating diverse inspiration sketches by integrating contextual semantic analysis. Additionally, the module incorporates an art information recommendation feature that suggests artist works, technique tutorials, or exhibition information related to the user's interests and learning path, further broadening their aesthetic horizons. This module not only spans the initial stage of inspiration generation but also continuously guides learners in creative thinking throughout subsequent creative practices. (See Figure 1)

4.2 Creative practice module

This module consists of two parts: the "Painting Step Generation System" and the "Multi-dimensional Recommendation System." The step generation system is trained on a large amount of real teaching data. For example, when students are creating sketches, the system follows the process of "composition and form establishment—painting shadows and dark areas—establishing major relationships—deepening details," intelligently generating reference images for the next steps based on semi-finished images uploaded by users, thus achieving a layered guidance function to assist in teaching. The recommendation system evaluates and guides works from four core dimensions: composition, proportion, structure, and

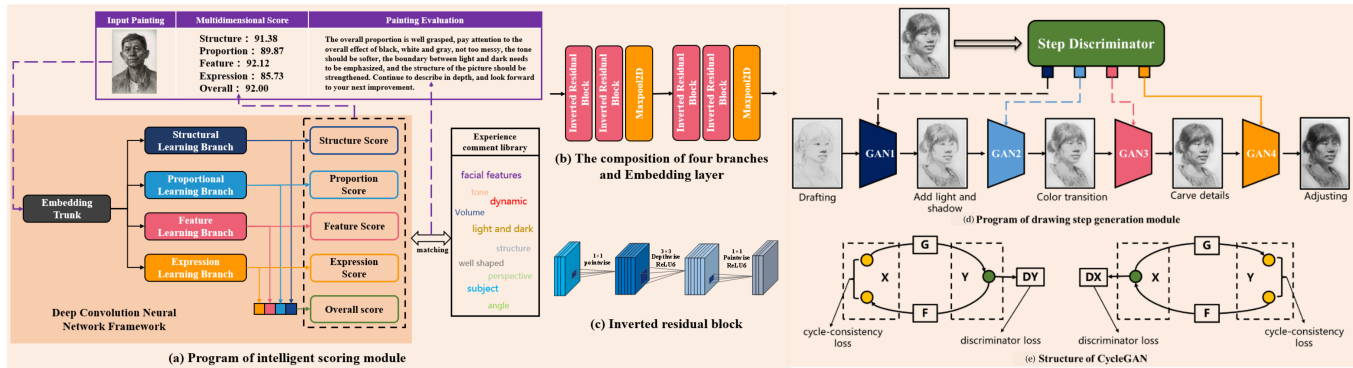


Figure 2: The architecture of scoring module. (a) Program of intelligent scoring module. (b) The composition of four branches and Embedding layer. (c) Inverted residual block.

lighting, helping students think creatively and track their learning progress. The collaboration between these two subsystems not only enhances students' self-learning abilities but also provides continuous, repeatable, and personalized "AI teacher" support for students in resource-poor areas, effectively alleviating the disparities in teaching caused by uneven distribution of teachers.

4.3 Sharing and communication module

To promote resource sharing and learning exchanges among students, the platform has designed a collaboration module centered on "Image Library Display-Tutorial Sharing-Personal Management," aiming to build an open and interactive aesthetic education community environment. This module consists of two main parts: the Image Library and the Tutorial Resource Center. The "Image Library" integrates high-quality works from various art categories for users to browse, reference, and copy online; the "Tutorials" section gathers a large number of high-definition sketch e-books and professional teachers' painting teaching videos, covering key skills such as basic composition, light and shadow handling, and detail rendering, categorized by art type for quick search and learning according to user needs. Additionally, under the "My" entry, the platform sets up a personal center section with functions including "Technique Changes," "Generation History," "Favorites," and "Edit Profile." The "Favorites" function allows users to save interesting image library works or tutorial resources to their personal accounts for easy future reference; "Generation History" and "Rating Records" meticulously document each AI-assisted generation and evaluation process, helping users review their creative paths, identify issues, and track progress. Through this mechanism, the platform has transformed from "unidirectional learning" to "multi-dimensional interaction," enhancing students' participation and motivation for continuous learning.

5 Technical Implementation

5.1 Multi-dimensional Visual Analysis Model for 2D Artworks

In this paper, we design a lightweight multi-dimensional scoring network based on the inverted residual block structure of MobileNetV2

[7]], which employs depthwise separable convolutions and a bottleneck design to improve computational efficiency. As shown in Figure 2c, this structure consists of an expansional convolution (1x1 convolution), separable convolutions (3x3 or 5x5), and a projection convolution (1x1). The expansional convolution is used to increase feature dimensions, the separable convolutions extract spatial features, and the projection convolution compresses channels to support residual connections.

The proposed scoring model structure is shown in Figure 2a, consisting of a graph embedding backbone and four scoring branches. The backbone maps the input sketch image to an appropriate feature space and extracts shared low-dimensional features; these features are then fed into four branches for specialized learning: feature, shape proportion, structural perspective, and detail effect. Each branch is followed by a linear projection layer that outputs scores corresponding to the dimensions, and the four-dimensional features are fused through another linear layer to predict the total score. Finally, by combining the scores from each dimension with a library of evaluation statements, a review of the work is generated. The network structure is shown in Figure 2b, which consists of four inverted residual blocks and two maximum pooling layers. The maximum pooling is introduced to enhance the generalization ability and accelerate the calculation, and batch normalization (BN) and ReLU activation function are added to improve the performance and alleviate the gradient disappearance problem.

5.2 Step-by-Step Guidance Model for 2D Artworks

In this paper, we use CYCLEGAN to build a step-generation network for artworks. CYCLEGAN is an effective variant of GAN that can transform images from one domain into another without matching data, generating new samples with high quality [3]. This network consists of a generator and a discriminator, which learn the data distribution through mutual competition. For the four steps in the student's painting process—composition and form, dark areas and shadows, establishing major relationships, and detailed rendering—we aim to have the model generate the next stage based on the previous one. To achieve this, we trained three CYCLEGAN-based step generators (as shown in Figure 2b), each with different

generative capabilities due to varying input data and parameters. Additionally, we trained a three-layer convolutional neural network as a step classifier (with kernel sizes of 3×3 and channel numbers of 32, 64, and 128) to automatically determine the current image's step and activate the corresponding generator to produce the next step image. The entire model was trained using 22,115 collected sketch portrait works, ensuring an equal number of works for each step. The training set accounts for 80%, and the test set accounts for 20%. All images were uniformly resized to $368 \times 256 \times 3$. The model is based on PYTHON 3.8 and PYTORCH 1.7.0 development, training on RTX3090 GPU, BATCH_SIZE set to 16, using ADAM optimizer, total training rounds are 200, initial learning rate is 0.0002, learning rate decreases by 0.1 every 50 rounds. Cycle consistency loss is used in the training process.

6 Conclusion

This study introduces an intelligent painting system designed to support sketch learning through multi-dimensional evaluation and step-by-step guidance. User testing and feedback confirm the system's effectiveness in enhancing students' drawing skills, particularly for those with intermediate abilities. By providing accessible and personalized artistic guidance, the system aims to promote educational equity, offering high-quality resources to learners regardless of their geographical location or institutional access. Future work will focus on expanding the system's capabilities and conducting broader studies to further validate its impact on diverse user groups.

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